



Diagnostic Study of Monsoon-Linked Rainfall Events: Unraveling the Role of Monsoon Low Pressure Systems across the Indian Subcontinent

Nitin Lohan^{1,2} · D. R. Pattanaik¹ · Sushil Kumar² · R. K. Giri¹

Received: 30 July 2024 / Accepted: 12 June 2025

© King Abdulaziz University and Springer Nature Switzerland AG 2025

Abstract

Monsoon systems play a pivotal role in shaping the hydrological landscape of the Indian Subcontinent, influencing the livelihoods of over a billion people through their profound impact on rainfall distribution. This study is conducted on an extensive assessment of extreme precipitation, precipitation trends, and its influencing factors spanning 41 years (1979–2019) within these monsoon systems, utilizing fifth-generation European Centre for Medium-Range Weather Forecasts (ECMWF) atmospheric reanalysis (ERA5). This comprehensive diagnostic study aims to understand the complex mechanisms underlying Monsoon Low-Pressure Systems (MLPSs) and how they are directly related to rainfall events associated with the monsoon. The paper outlines the complex function of these monsoon systems in regulating intense precipitation occurrences in the Indian Subcontinent. The study looks into the main variables influencing MLPSs-related changes in rainfall patterns. It aims to unravel the complex relationship between these systems and the erratic but essential precipitation events that they set off. With the study of 252 monsoon depressions and 305 monsoon lows, our findings reveal distinct trends in rain rates within specific quadrants of monsoon depressions and lows. Notably, there has been an increase in precipitation rates observed in the southwest quadrant of the vortex center. In contrast, the eastern quadrants experienced a decline in precipitation in case of monsoon depression. However, in the case of lows, this pattern is different. This study also analyzes the influencing factors associated with rainfall and their correlation with rainfall.

Graphical Abstract

This graphical abstract visually summarizes a 41-year (1979–2019) diagnostic study on monsoon low-pressure systems (MLPSs) and their influence on rainfall across the Indian Subcontinent, utilizing ERA5 reanalysis data. The left panel displays a hexbin map of monsoon depressions, and the right panel shows monsoon lows, indicating the frequency of occurrence. Six subplots between the maps depict vortex-centered rainfall composites (mm/day) for monsoon depressions and lows across June, July, August, and September (JJAS). For monsoon depressions, precipitation rates in the southwest quadrant increased by 6 mm/day/decade (95% confidence level), while northern sectors show a slight decline (not significant). Monsoon lows exhibit a smaller increase of 3 mm/day/decade in the northwest quadrant, with a minor decrease in the southwest (not significant). The bottom panels present a time series of precipitation rates with red and blue lines showing annual variability and trends, respectively, with a yellow text box noting an increase in precipitation rates in the southwest quadrant of the vortex center. In this core finding, depressions show a significant upward trend and lows a broader but less intense pattern. Here, upward arrows denote increasing trends and downward arrows denote decreasing trends in frequency and intensity of these systems.

✉ D. R. Pattanaik
drpattanaik@gmail.com

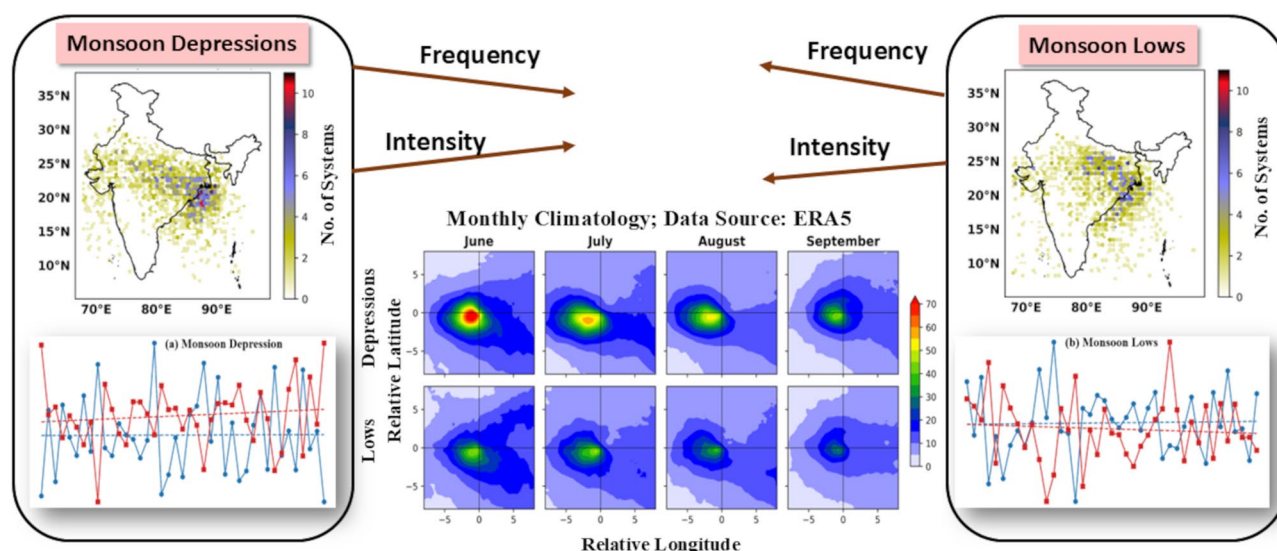
Nitin Lohan
lohannitin@gmail.com

Sushil Kumar
sushil12@gmail.com

R. K. Giri
rk.giriccs@gmail.com

¹ India Meteorological Department (IMD), Ministry of Earth Sciences (MoES), Lodi Road, New Delhi 110003, India

² Department of Applied Mathematics, Gautam Buddha University, USoVSAS, Greater Noida 201312, India



Keywords Monsoon depression · Monsoon low pressure · Indian monsoon · Rainfall · Vorticity advection

1 Introduction

The synoptic-scale cyclonic disturbances, often known as Monsoon Low-Pressure Systems (MLPSs), develop throughout the Indian subcontinent during the summer monsoon season, June to September (Rao 1961; Mooley and Shukla 1989; Hurley and Boos 2015). The Indian summer monsoon season months are June, July, August, and September (JJAS). This monsoon system contributes approximately 80% to the annual precipitation in the Indian subcontinent (Parthasarathy et al. 1994). These weather systems, roughly divided into more substantial and weaker parts known as Monsoon Depressions and lows, play a crucial role in the regional climate (Thomas et al. 2023). On average, Indian MLPSs usually last about 3 to 6 days, and typically, there are approximately 14 occurrences of LPSs during the summer season (JJAS) (Mooley and Shukla 1989; Hurley and Boos 2015). Most of these MLPSs originate over the warm waters of the north Bay of Bengal (BoB) and follow a westward or northwestward path along the monsoon trough (Ding and Sikka 2006; Mohapatra and Mohanty 2009). Within this densely populated region, instances of monsoon precipitation can be highly concentrated temporally, leading to substantial rainfall—sometimes exceeding 15 cm within a single day over extensive areas encompassing central India, the Himalayas, and India's west coast (Vishnu et al. 2023). The study of rainfall events associated with monsoon depressions over the Indian region is paramount, considering the reliance of over one billion people on water supplied by the South Asian monsoon.

Multiple studies have tried to understand how monsoon depressions develop and the overall workings of MLPSs. Early investigations by Sikka (1977) and Keshavamurty et al. (1978) initially proposed several mechanisms for the growth of monsoon Low-Pressure Systems (LPS), including barotropic and baroclinic instabilities, as well as their interaction with moist convection. The relative divergence between 1000 hPa and 700 hPa from contour configuration plays a vital role in these systems (Sawyer and Matthewman 1951). Goswami (1987) suggested an alternative view, positing that these systems are Rossby-gravity waves propagating northwestward due to the coupling between cumulus convection and low-level cyclonic vorticity maxima. According to Bhalme (1972), there has been a noticeable decline in cyclones and depressions throughout the southwest monsoon season since the turn of the 19th century. This study was done through the period of 1891 to 1970. A satellite study determined that the centers were displaced to the southwest with respect to isobaric centers in most cases (Keshavamurty 1972).

Several studies (Hahn and Manabe 1975; Chakraborty et al. 2002, 2006; Zhang et al. 2015) have extensively explored the influence of orographic barriers on the Indian monsoon's circulation and precipitation patterns. (Nie et al. 2010) and (Boos and Kuang 2010) argued that the South Asian summer monsoon system operates under convective quasi-equilibrium. They proposed that local moist energy sources and orographic effects prevent dry intrusions, maintaining elevated sub-cloud-layer moist entropy over continental India and sustaining robust monsoon circulation.

Although the relationship between MLPs and extreme precipitation occurrences has been the subject of numerous studies (Pattanaik and Rajeevan 2010; Ajayamohan et al. 2010; Nikumbh et al. 2020; Thomas et al. 2021). (Hunt and Fletcher 2019) studied the effective radius of Monsoon Depression and MLPs. Vishnu et al. (2023) stated that the rainfall associated with monsoon depression showed an increasing trend in the southwest sector of monsoon depression. The rainfall belt over the Indian subcontinent has shifted eastward intermittently, and this change is particularly pronounced during the withdrawal phase (Pattanaik 2007).

Recently, (Vishnu et al. 2020) developed an hourly dataset for the tracks and intensities of low-pressure systems (LPS) in the global tropics (35°S–35°N) for five atmospheric reanalyses. This current study increases the work of these recent studies by defining the climatology of rainfall, vorticity advection, vertically integrated moisture transport, and all other variables that influence rainfall during monsoon. This study includes an analysis of 252 monsoon depression and 305 monsoon lows. This study also includes the rainfall trend and other variables around monsoon depression and lows center. This study also analyses the rainfall pattern associated with monsoon depression and lows during the El Niño–Southern Oscillation (ENSO), El Niño and La Niña periods. The structure of the paper is as follows: In Sect. 2, the study's data and methodology are described. Section 3 describes the results and discussion, and Sect. 4 concludes this article.

2 Data and Methodology

2.1 Datasets

2.1.1 ERA5

The European Centre for Medium-Range Weather Forecasts (ECMWF), supported by 34 countries, is the international authority in weather forecasting research and operations. Their latest advancement, ERA5, represents the fifth-generation reanalysis surpassing its predecessor, ERA-Interim. With its extensive collection of meteorological data from 1940 to the present, ERA5 provides hourly estimates of atmospheric variables at a 31 km horizontal resolution across 137 pressure levels. With the use of a large variety of historical observations—especially satellite data—and intricate data assimilation and modeling algorithms, ERA5 generates more precise forecasts of atmospheric conditions (Hersbach et al. 2020). This research explicitly employs ERA5's daily data presented on standardized latitude-longitude grids at a resolution of $0.25^\circ \times 0.25^\circ$ from 1979 to

2018. Data is available at: <https://cds.climate.copernicus.eu/#/home>.

ERA5 reanalysis data is chosen for this study due to its superior spatial and temporal resolution and enhanced data assimilation techniques, ensuring the most accurate representation of monsoon dynamics.

2.1.2 Monsoon Depression Tracks Data

The global track dataset of monsoon low-pressure systems encompasses the tracks and intensities of low-pressure systems (LPS) within the tropical belt (35°S–35°N), identified across five atmospheric reanalyses, including ERA5. Vishnu et al. (2020) compiled this dataset using an optimized tracking algorithm, employing an automated Lagrangian pointwise feature tracker for precisely identifying and monitoring these systems. Data is available at: <https://zenodo.org/records/3890646#.ZAmsli9U1QI>.

The data is divided into monsoon depressions and monsoon lows. The India Meteorological Department (IMD) traditionally categorized these storms by intensity, with the weakest systems called monsoon lows (surface wind speeds less than 8.5 m/s and mean sea level pressure (MSLP) at least 2 hPa lower than surrounding regions), stronger systems called monsoon depressions (wind speeds 8.5–13.5 m/s and MSLP anomalies 4–8 hPa), and even stronger vortices called deep depressions and cyclonic storms.

2.1.3 Classification of El Niño, La Niña Years

El Niño conditions persist when the average six-month running means of sea surface temperature (SST) anomalies within the Niño 3.4 region of the equatorial Eastern Pacific Ocean (5°N–5°S, 120°E–180°W) exceed 0.4°C (Bhatla et al. 2020). The Niño 3.4 SST index, derived from observed data from 1979 to 2018 and sourced from NOAA's Climate Prediction Centre (www.psl.noaa.gov/data), is the basis for this classification. Among the 40 years analyzed, eight years are categorized as El Niño, while eleven years are identified as La Niña, leaving the remaining twenty-one years classified as neutral years (Table 1).

2.2 Methodology

The methodology employed in this study involved the utilization of multiple datasets and computational techniques to generate comprehensive analyses of the impacts of monsoon depressions and low-pressure systems on rainfall patterns and atmospheric dynamics. The primary data sources encompassed the ERA5 dataset for atmospheric variables such as rainfall, wind, moisture transport, and vorticity, along with a text file derived from the tracks dataset

Table 1 List of El Nino and La Nina events during 1979–2019

El Nino Years	La Nina Years
1982	1984
1987	1985
1991	1988
1997	1995
2002	1999
2004	2000
2009	2007
2015	2008
	2010
	2011

containing specific latitude, longitude, and temporal information for these meteorological phenomena. To construct the rainfall composite, each recorded latitude, longitude, and timesteps from the tracks dataset was cross-referenced with the ERA5 dataset, extracting corresponding atmospheric data within a 8-degree square region centered on the given coordinates. Mean rainfall values were computed within these areas, establishing a composite representation of rainfall patterns associated with the identified monsoon depressions and low-pressure systems. This method is the same for all composite plots.

Vertical integrated moisture flux Q can be expressed as

$$Q = \frac{1}{g} \int_{p_s}^{300} qV dp \quad (\text{Ullah and Gao 2012; Lohan et al. 2025})$$

where g is the acceleration due to gravity, p_s is surface pressure, V is wind vector and q is specific humidity. For vertical integrated moisture transport (VIMT), vertical integration of this equation is carried out from surface pressure in hPa to 300 hPa since specific humidity has a negligible effect above this level and is not a part of reanalysis (Kalnay et al. 1996).

Also, the vorticity advection is calculated as:

$$\text{Vorticity advection} = -v \frac{d\zeta}{dy} - u \frac{d\zeta}{dx} \quad (1)$$

where u and v are zonal and meridional wind components, and

$\frac{d\zeta}{dx}$, $\frac{d\zeta}{dy}$ denotes the differentiation of relative vorticity concerning longitude and latitude, respectively.

The trend analysis is done with the Mann-Kandall (MK) Test. The MK test is used as:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{sgn}(x_j - x_i)$$

where n is the number of points in data. x_j and x_i are values of data and time j and i ($j > i$). $\text{sgn}(x_j - x_i)$ is the sign function, which is 1 if $(x_j - x_i) > 0$, 0 if $(x_j - x_i) = 0$ and -1 if $(x_j - x_i) < 0$. For large samples ($n > 8$), S is approximately normally distributed, with the mean (Mondal et al. 2012).

2.3 Study Area

For this study, a focus has been placed on low-pressure systems originating within a defined range between $5^\circ N$ to $40^\circ N$ and $65^\circ E$ to $100^\circ E$. This chosen region is shown on a map in Fig. 1. This figure also provides information on the occurrence and distribution of these low-pressure systems throughout the designated geographic area by offering how frequently they pass through various locations. The study's temporal scope is from 1979 to 2019, allowing for collecting and analyzing these low-pressure systems' patterns and behaviors throughout forty years.

3 Results and Discussion

3.1 Monsoon Depressions and Lows Tracks

A hexbin representation of monsoon depressions and monsoon lows tracks showing the frequency of occurrence over the study region from 1979 to 2019 is shown in Fig. 1(a-b). Darker colors represent a higher frequency of occurrences, and each hexagonal bin represents the number of tracks recorded across the corresponding geographic area. These tracks are concentrated in a trail that crosses the eastern and central parts of the mainland and BoB, indicating a solid monsoon activity corridor. The distribution of these tracks is pivotal in understanding the climatology of monsoon systems, as areas with a higher frequency of passages correlate with increased rainfall events and potential impacts on the region's hydrology and agriculture.

3.2 Trend Analysis of Monsoon Depressions and Monsoon Lows

In the context of the changing global climate, understanding the variability and shifts in regional monsoon systems is crucial for predicting socio-economic impacts (Lal et al. 2001). This section provides a comprehensive analysis of the trends in the frequency, geographical shifts, and intensities of monsoon depressions and monsoon lows. The frequency analysis of monsoon depression (Fig. 2a) and lows (Fig. 2b) indicates the variability of these systems over the past four decades. Figure 2a shows a slightly decreasing trend in the annual count of monsoon depressions. In

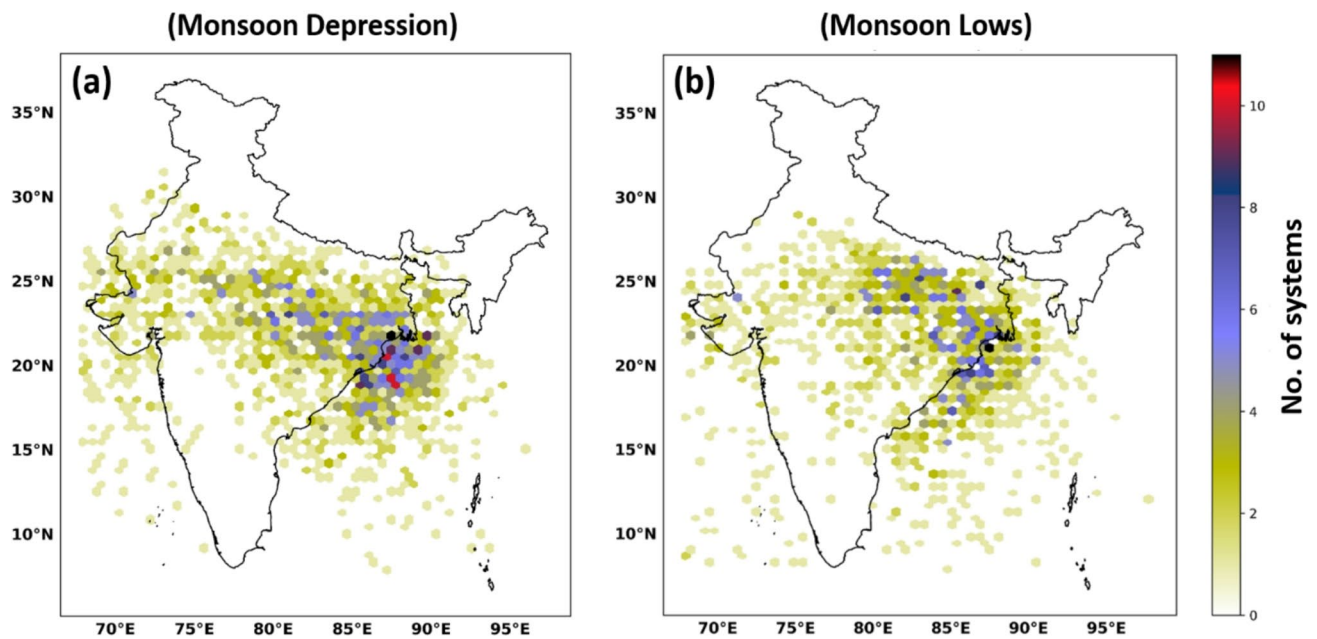


Fig. 1 The area selected for this study and the spatial distribution of (a) Monsoon Depressions and (b) Monsoon Lows tracks from 1979 to 2019. The colored hexbins represent the number of systems observed at each location across India

contrast, the frequency of lows shows a slightly increasing trend though the statistical significance remains weak. (Fig. 2b). These trends suggest a potential shift in the climatic factors that drive these systems, such as sea surface temperature and atmospheric circulation pattern. The study does not explicitly calculate a statistical correlation coefficient between the annual number of monsoon lows and monsoon depressions over the 1979–2019 period. These trends in monsoon depressions and lows are similar to many previous studies (Dash et al. 2004; Mohapatra et al. 2017).

The analysis of geographical shifts (Fig. 2c) reveals minute yet significant movements in the spatial centers of these systems. Monsoon depressions demonstrate a gradual westward shift in the mean longitude (Fig. 2e) with a very small change in the mean latitude position (Fig. 2c). Conversely, the mean longitude of monsoon lows shows little to no significant trend, while the mean latitude of monsoon lows shows a shift of location to northward. These changes may be due to the dominant wind patterns and oceanic conditions in the region. These shifts could lead to changes in rainfall patterns around these systems.

Figure 3 illustrates the interannual variability and trends in the intensity of monsoon depression and monsoon lows from 1979 to 2019. The intensity of these systems is defined through mean sea level pressure (MSLP) and maximum sustained wind (MSW). From Fig. 3a–b, it is clear that the trend in MSLP in both monsoon systems is negligible, but the trend in MSW is notable. The trend in MSW is found to be increasing in case of monsoon depression, but the reverse trend in case of monsoon lows. However, all these trends

lack statistical significance. These significance are assessed using the Mann-Kendall (MK) test (Sect. 2.2), which yielded p -values greater than 0.05 ($p \approx 0.12$ for depressions, $p \approx 0.15$ for lows), indicating that we cannot reject the null hypothesis of no trend at the 95% confidence level. The intensities could be attributed to the warming ocean and altered wind patterns. These findings are crucial for understanding the changes in monsoon systems.

3.3 Precipitation Analysis

This section provides an analysis of the climatological rainfall of these monsoon systems and the trend of rainfall around the center of the system. The analysis of rainfall is based on the monsoon depression center at 00UTC, with rainfall data representing the accumulated precipitation over the subsequent 24-hour period. The spatial distribution and intensity of rainfall associated with monsoon depressions and lows, utilizing composite plots derived from ERA5 meteorological data spanning four decades (1979 to 2019), is shown in Fig. 4a–b. The analysis discerns the rainfall characteristics relative to the center of these weather systems. Figure 4a shows that monsoon depressions are characterized by a highly concentrated core of intense rainfall exceeding 50 mm/day, diminishing outwardly but notably more pronounced in the southwestern quadrant. This sectoral disposition suggests a significant directional influence on the distribution of intense precipitation.

On the other hand, Fig. 4b illustrates that monsoon lows, despite a lower peak intensity near 35 mm/day, exhibit a

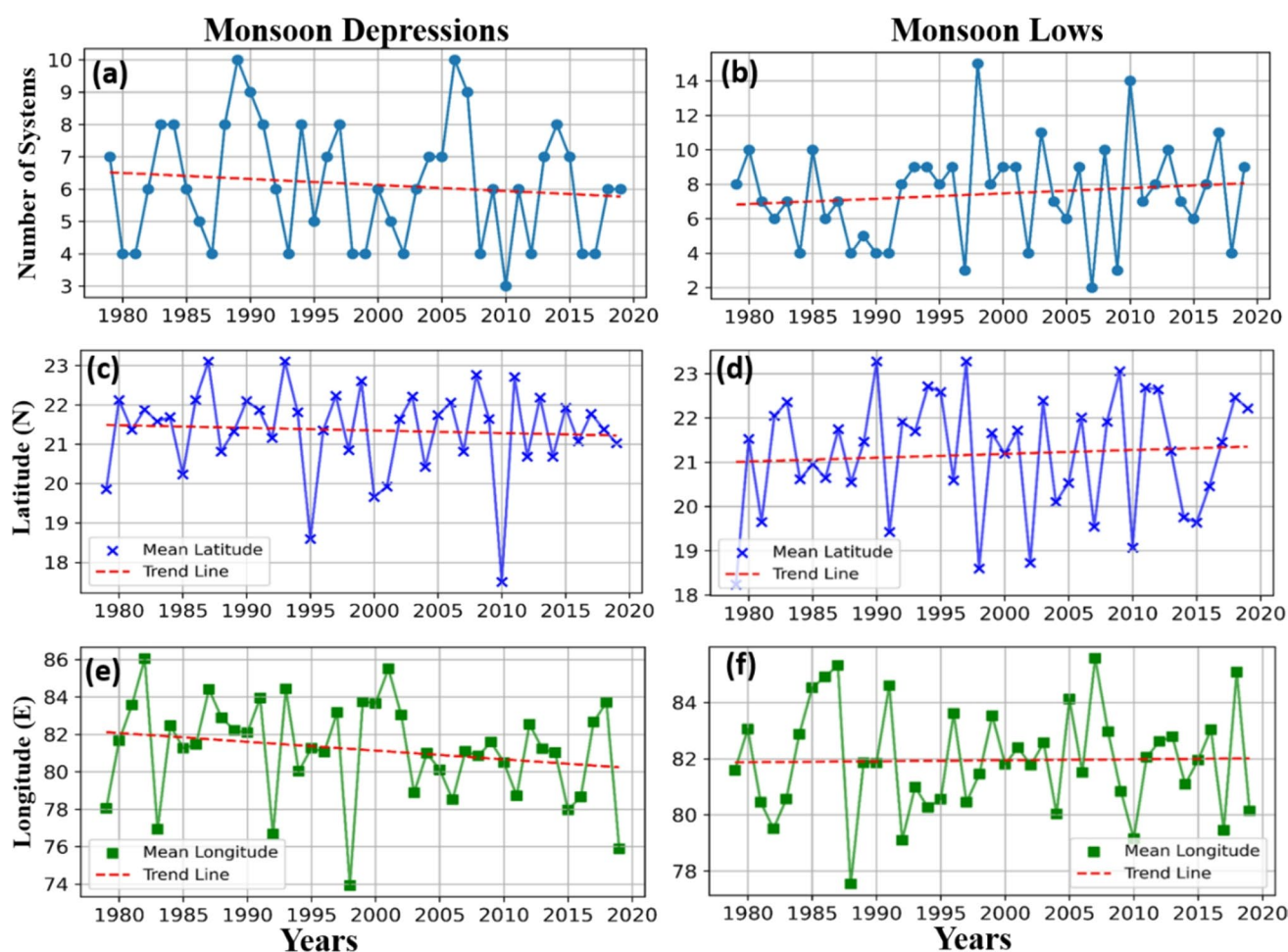


Fig. 2 Temporal Analysis of Monsoon Depressions and Monsoon Lows from 1979 to 2019 The figure shows trends and variations in (a) the number of monsoon depressions per year, (b) the number of monsoon lows per year, (c) mean latitude of monsoon depressions,

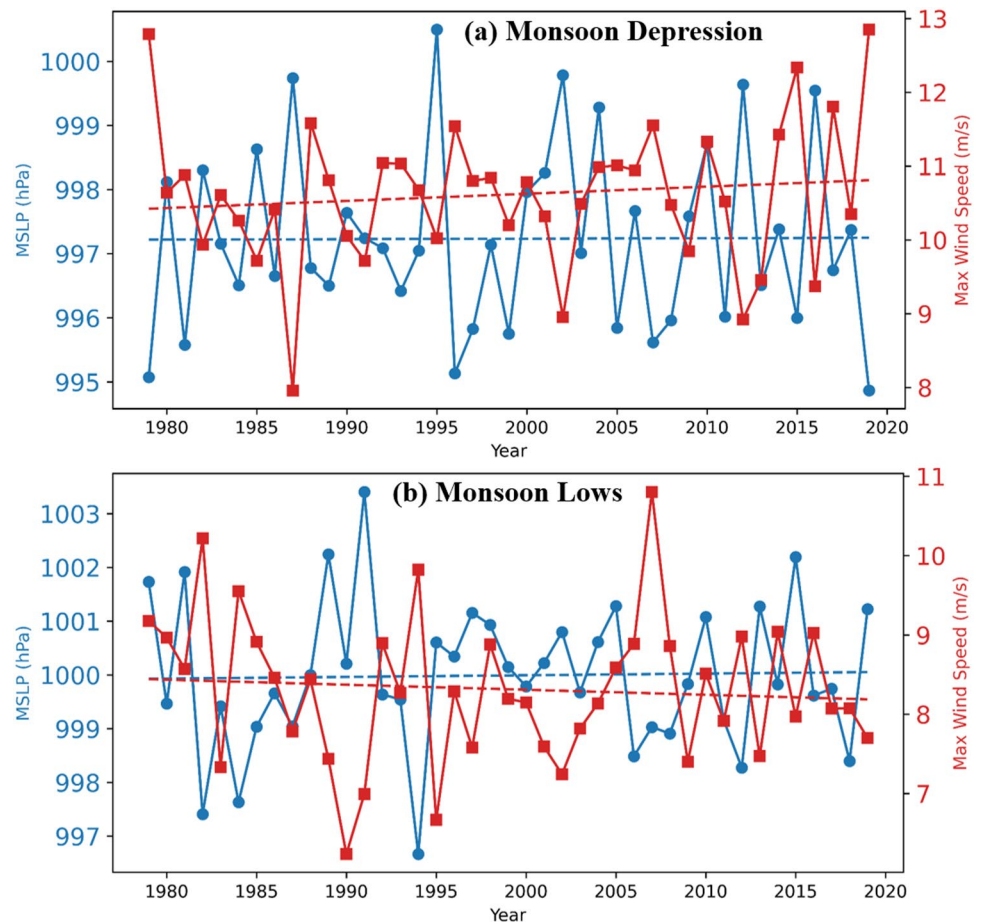
(d) mean latitude of monsoon lows, (e) mean longitude of monsoon depressions, (f) mean longitude of monsoon lows. The markers indicate the annual counts or average positions, and the dashed red line shows the trends

wider expanse of rainfall distribution. However, similar to monsoon depressions, the southwest sector is primarily the most affected area, indicating a consistent pattern in rainfall disposition across both meteorological phenomena. However, the spatial extent of 10 mm/day rainfall is similar in both monsoon depression and lows in the selected 8°×8 degree box. The consistent focus on the southwest region in both systems could indicate a solid topographical or meteorological factor affecting this distribution. The shift in rainfall from eastern to southwestern regions may result from enhanced negative vorticity advection and moisture convergence in the southwest region (Sect. 3.6), potentially amplified by orographic influences and warming climate, which increase moisture availability. These observations distinguish the rainfall patterns of monsoon lows and depressions and draw attention to the possible consequences for regional climate adaptation plans and hydrological effects.

For monsoon depression, the trend analysis reveals a substantial increase in rainfall within the southeastern and

southwestern sectors with an intensification of more than 6 mm/day/decade (Fig. 4c). These significant upward trends in the specified sectors suggest amplifying monsoon depression-related rainfall, which may have critical hydrological implications. In contrast, for monsoon lows, the positive rainfall trends are more prominent in the northwestern sector, although the overall magnitude of change is smaller than in monsoon depressions (Fig. 4d). In addition, a slight decreasing trend in rainfall in the southern sector for monsoon depressions, while a similar decline in the southwestern sector for monsoon lows is also observed (Fig. 4c-d). This spatial trend difference highlights the distinct meteorological behaviors driving these monsoon systems and emphasizes the complexity of regional precipitation patterns influenced by the varying dynamics of monsoon depressions and lows. The dots across various regions denote statistical significance (95%), asserting higher confidence in the observed trends. The trend intensity and geographical distribution disparity between monsoon depressions and

Fig. 3 The Trends in Atmospheric Pressure and Wind Speed for Monsoon Systems from 1979 to 2019 **(a)** Variability and trends in mean sea level pressure (hPa) and maximum wind speed (m/s) associated with monsoon depressions. **(b)** Variability and trends in MSLP and maximum wind speed associated with monsoon lows. Each plot shows the annual values of MSLP (blue line) and maximum wind speed (red line) over the study period, with trend lines indicating changes over time



lows underscores the complex nature of regional precipitation changes and their sector-specific impacts. Such sectoral trends in rainfall are consequential for regional agricultural practices, water resource planning, and understanding the spatial dynamics of monsoon systems in the context of global climate variability.

3.4 Monthly Rainfall Variation during Monsoon Depressions and Lows

The monthly variations of rainfall associated with monsoon depressions and monsoon lows are shown in Fig. 5(a-h) for individual months of JJAS. In the monthly distribution analysis of monsoon depression characteristics, our methodology accounts for the fact that if a monsoon depression originates in June and ends in July, the days falling within June are included in the analysis for June. Similarly, the days in July are attributed to the July analysis. This allows for precisely assessing the monsoon's monthly progression and associated meteorological impacts. In June (Fig. 5a), the rainfall is markedly intense, with peak values reaching above 60 mm/day from monsoon depression, suggesting the monsoon onset phase typically brings substantial precipitation with monsoon depression. In June, the onset of

the South Asian monsoon begins. A dramatic increase in rainfall marks this period as moist ocean air is drawn into the subcontinent. Progressing into July (Fig. 5b), the core of heavy rainfall remains robust, although slightly less concentrated than in June. By July, the monsoon is well established. The core of heavy rain remains intense, supporting the crucial agricultural season and affecting daily life and the regional economy. However, the distribution might start showing slight shifts from the initial onset patterns, indicating the dynamic nature of the monsoon. Conversely, in case of monsoon lows, the rainfall pattern is almost the same but less intense (Fig. 5e).

The continuation into August (Fig. 5c, g) shows a similar pattern to July for both monsoon systems, with a high concentration of rainfall, indicative of the peak monsoon season. However, by September (Fig. 5d, h), there is a noticeable diminishment in rainfall intensity, with peak values decreasing significantly for both monsoon depressions and lows. This phase is critical as it indicates the withdrawal of monsoon. These analyses underscore the dynamic nature of monsoon depressions and their variable influence on monthly rainfall distribution.

Our analysis of monthly rainfall trends associated with monsoon depression and monsoon lows for JJAS are shown

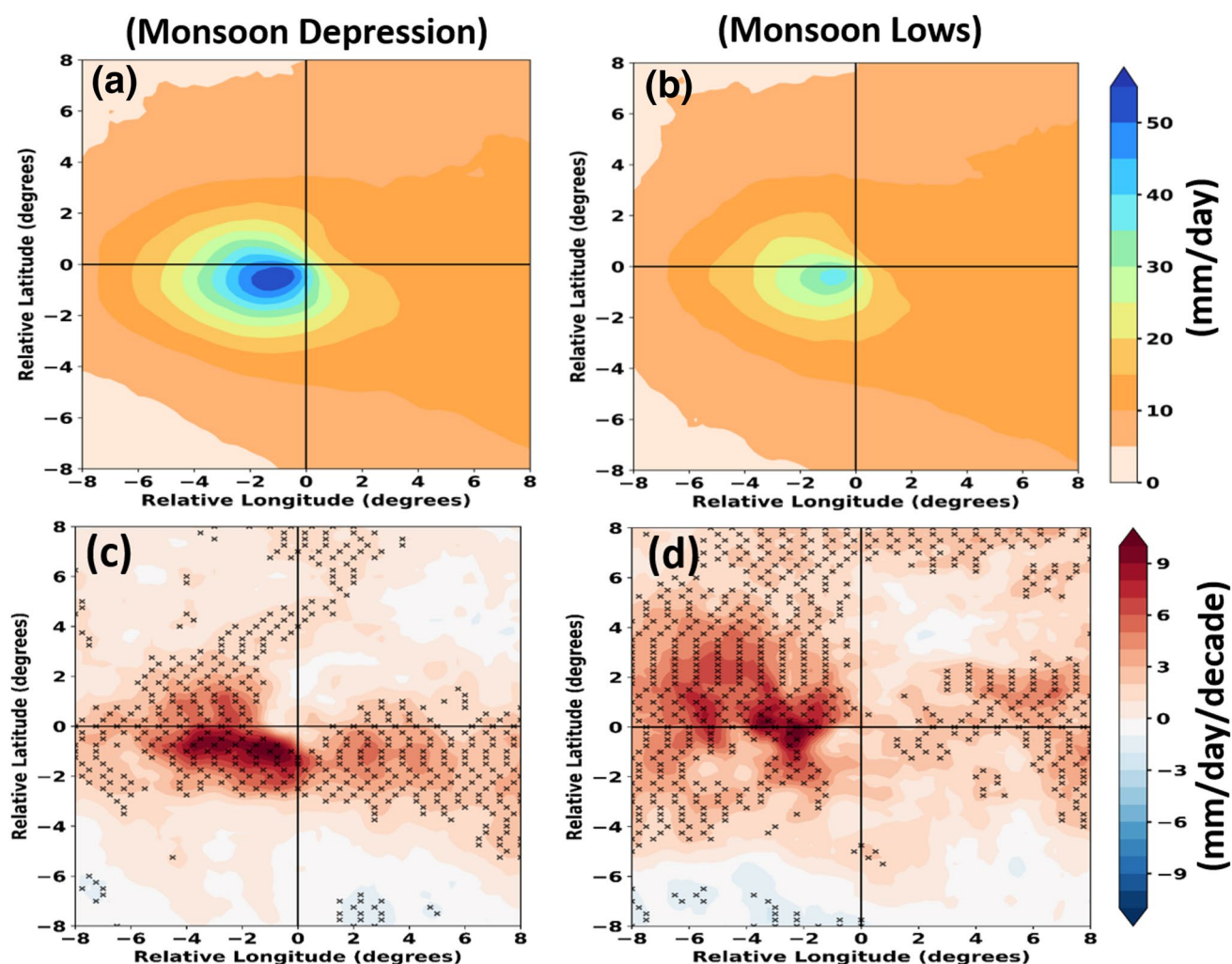


Fig. 4 Climatological Rainfall over the study period 1979–2019 for JJAS (mm/day) for (a) Monsoon Depressions, (b) Monsoon Lows around the center of Monsoon systems. The trend in rainfall over

decades (mm/day/decade) for (c) monsoon depressions and (d) monsoon lows, with statistical significance marked by crosses

in Fig. 6a. During June, monsoon depressions exhibit a significant increase in rainfall across the central and southwestern regions (Fig. 6a). This trend pattern remains similar in July month for monsoon depressions, indicating ongoing monsoon activity (Fig. 6b). However, as the season progresses into August, the rainfall pattern becomes more variable, though the central monsoon belt still shows a positive trend (Fig. 6c). In case of September the trend is again more positive, indicating late monsoon withdrawal phase (Fig. 6d).

In contrast, monsoon lows present a different pattern, with a decrease in rainfall noted across most regions in the month of June (Fig. 6e). This suggests either a shift in the dynamic of monsoon lows or a possible delay in their seasonal onset. This decreasing trend continues into July, especially pronounced in the southern sector (Fig. 6f). However, by August, some recovery is observed in central regions,

where rainfall increases, aligning with the typical behavior of monsoon lows during the peak of the monsoon (Fig. 6g). The trend for monsoon lows in September underscores a significant high in rainfall across a wide area, more pronounced than that observed for monsoon depressions, indicating a substantial shift in the behavior of these systems as the monsoon season draws to a close (Fig. 6h). The distinct monthly and spatial trends in rainfall from monsoon depressions and lows are indicative of their differential impacts on the seasonal monsoon dynamics.

3.5 Monsoon Depressions and Lows Associated Rainfall in Relation To ENSO

Figure 6a–d illustrates vortex-centered composites of rainfall (mm/day) during monsoon depressions and monsoon lows throughout 1979–2019, with (a) and (b) illustrating the

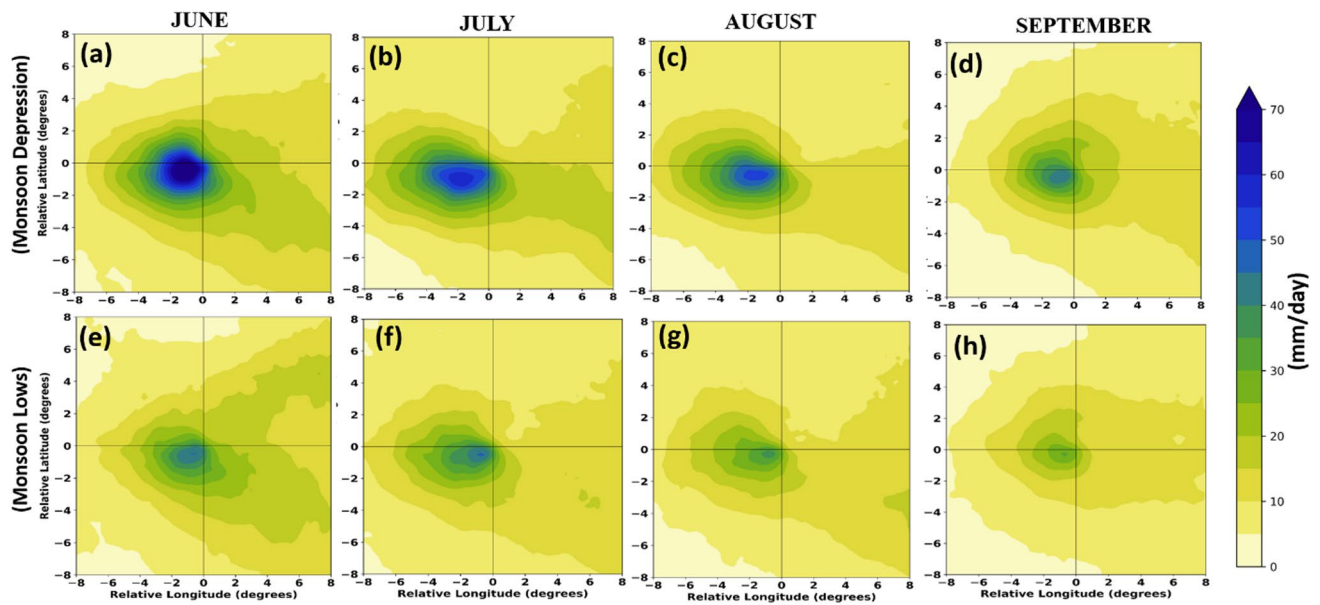


Fig. 5 Monthly climatological rainfall patterns (mm/day) over the study period 1979–2019 for Monsoon Depressions (a–d) and monsoon lows (e–h). The respective months are written on the top of the plot

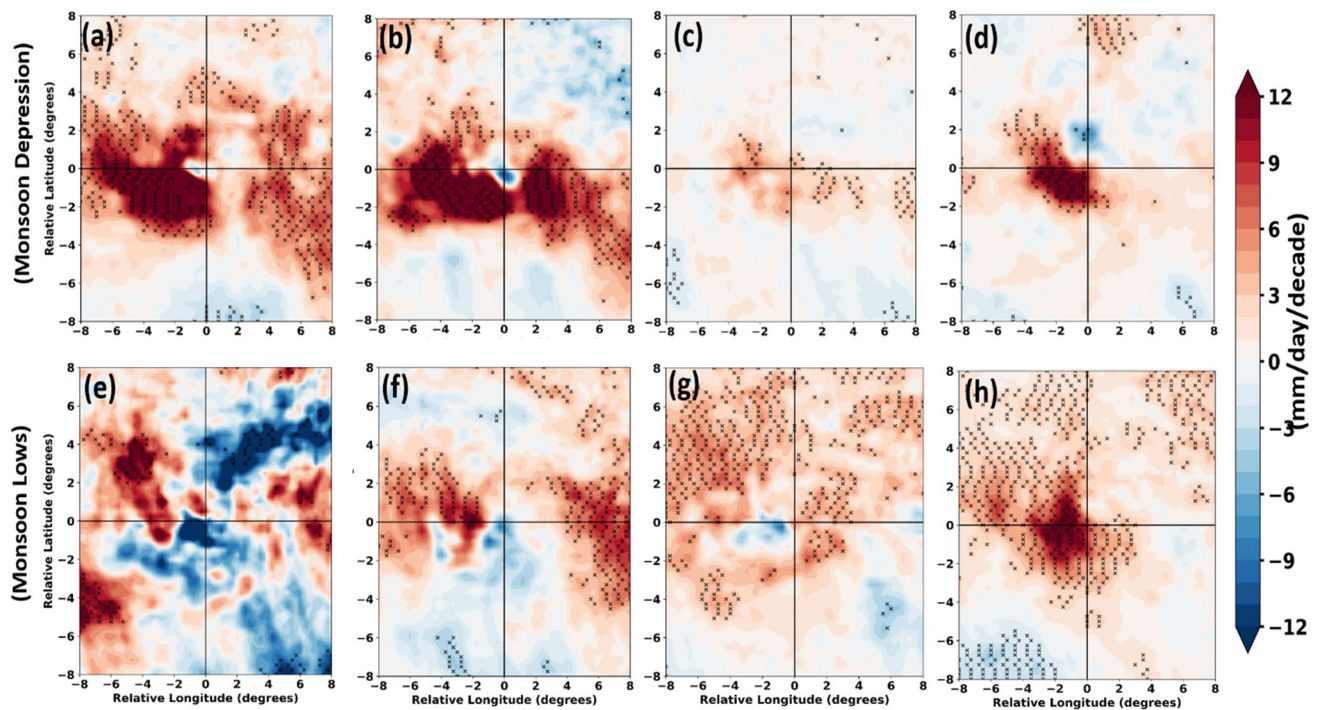


Fig. 6 Same as Fig. 5 but for trends in monthly rainfall (mm/day/decade)

rainfall distribution during La Niña and El Niño for monsoon depressions and (c) and (d) representing the same for monsoon lows. The events' corresponding years are shown in Table 1. In La Niña conditions (Fig. 6a), the rainfall is more intense and widespread, with a peak well above 65 mm/day, indicating a significant precipitation enhancement within the central region surrounding the monsoon

depression. This suggests a strong coupling between La Niña conditions and the intensification of monsoon rainfall. On the other hand, during El Niño events (Fig. 6b), while the peak rainfall remains considerable, there is a noticeable reduction in overall intensity and spatial coverage, with maximum values reaching around 45–50 mm/day. Similarly, monsoon lows during La Niña (Fig. 7c) also show an

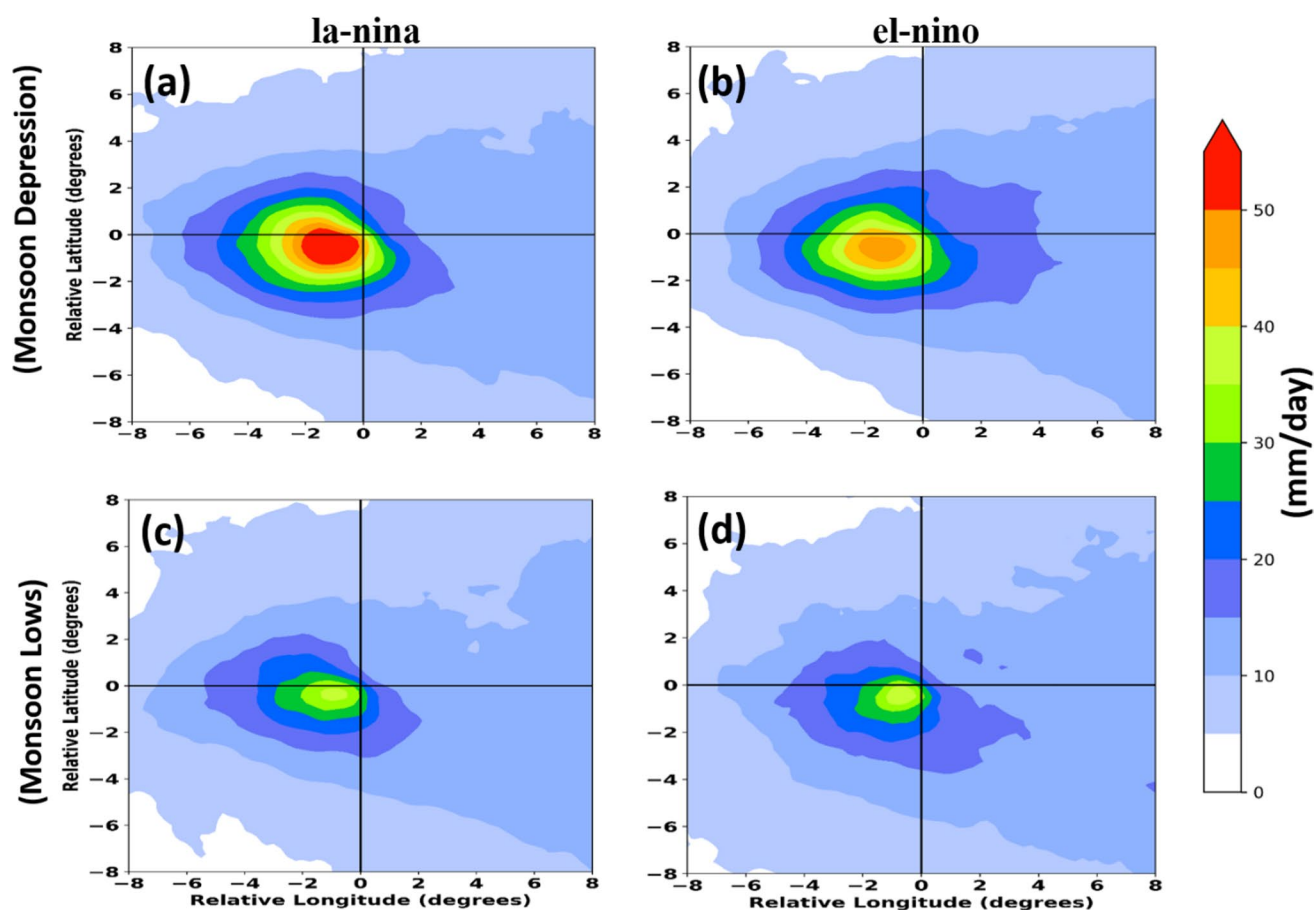


Fig. 7 Vortex-centered composites of rainfall (mm/day) for (a) rainfall during La Niña events associated with monsoon depressions, (b) rainfall during El Niño events associated with monsoon depressions, (c)

rainfall during La Niña events associated with monsoon lows, and (d) rainfall during El Niño events associated with monsoon lows

increase in rainfall, though less pronounced than in depressions, suggesting a moderate yet significant influence of La Niña on these weaker monsoon systems. In contrast, under El Niño conditions (Fig. 7d), monsoon lows experience an even more substantial decrease in rainfall compared to depressions, highlighting a heightened sensitivity of monsoon lows to El Niño suppressive impacts. This pattern suggests that El Niño conditions are associated with a relative suppression of rainfall in monsoon systems. The enhanced rainfall during La Niña likely results from a strengthened Walker circulation, increasing moisture convergence over the Indian Subcontinent, while El Niño's warmer SSTs weaken this circulation, reducing monsoon rainfall (Bhatla et al. 2020). These distinct composites highlight the contrasting impacts of ENSO phases on monsoon systems characteristics, which can be critical for understanding regional hydrological responses and informing water management strategies in the face of climate variability.

3.6 Dynamical Parameters Associated with Monsoon Depressions and Lows

A comprehensive analysis of associated meteorological parameters is essential to prove the observed rainfall patterns linked with monsoon depressions and monsoon lows. This includes examining vorticity advection at 850 hPa, vertically integrated moisture convergence (VIMC) from surface to 300 hPa, convergence of airflow, and prevailing wind patterns at 850 hPa. These variables provide a composite perspective on the dynamics during monsoon depressions. Each parameter is scrutinized concurrent with the time of the monsoon depression center, utilizing hourly data. The following sections will present detailed analyses of these variables, focusing exclusively on events of monsoon depressions to examine their contributory roles in influencing the monsoon's behavior and resultant precipitation patterns.

Figure 8a and d represent the vorticity advection patterns around monsoon depression centers during the monsoon months (JJAS). Vorticity advection, which involves

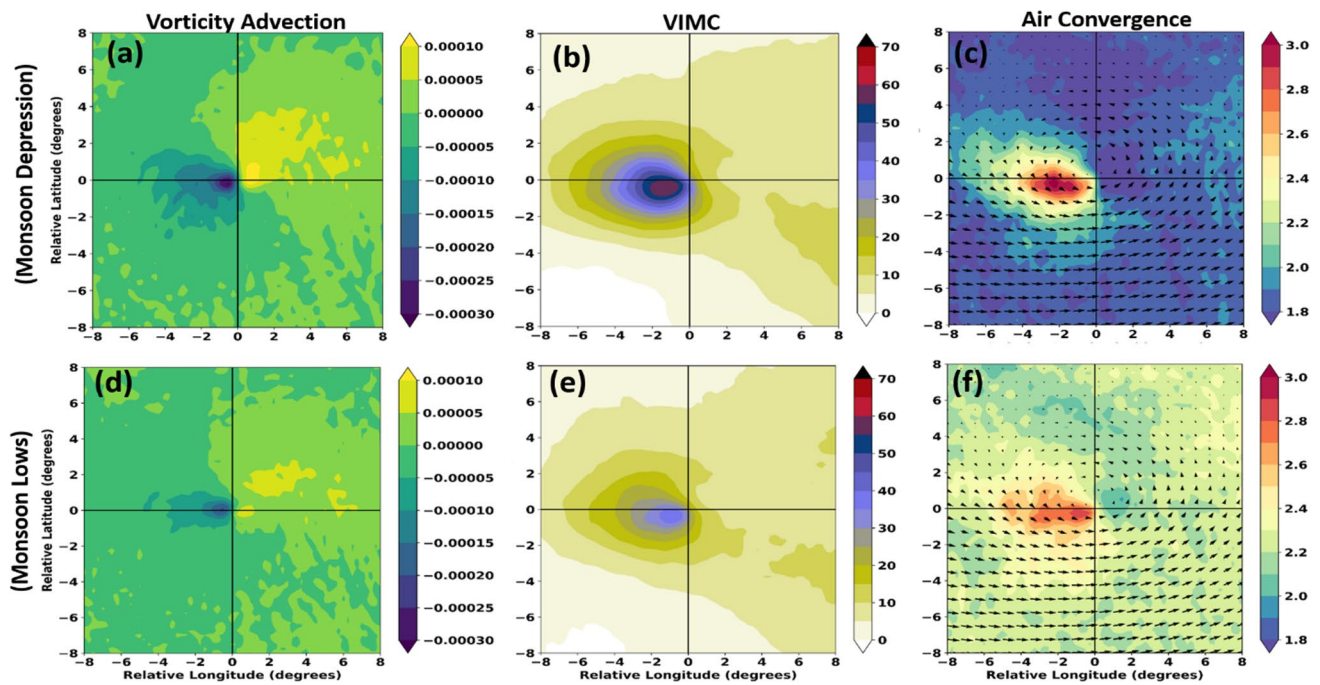


Fig. 8 Vortex-centered composites of (a) vorticity advection at 850 hPa, (b) mean vertically integrated moisture convergence (kg/m^2) from surface to 300 hPa, (c) air convergence associated with monsoon depressions (s^{-1}) at 850 hPa, and (d–f) are the same as (a–c) but for monsoon lows

the transport of vorticity by the wind, is crucial for understanding cyclonic activity. For monsoon depressions, negative vorticity advection is prevalent in the southwestern sector, indicative of enhanced atmospheric convergence in this region. This convergence is typically associated with upward motion and cloud formation, which can lead to increased precipitation (Martin 2014). In contrast, monsoon lows, which generally exhibit weaker vorticity advection, correspond to less intense rainfall patterns, reflecting their typically weaker cyclonic characteristics (Fig. 8d). The spatial correlation between the higher negative vorticity advection and the observed increased rainfall in the same southwestern sector suggests a coherent relationship between these meteorological variables (Fig. 9a and d). On the other hand, positive vorticity advection corresponds to areas of divergence, which are generally less favorable for precipitation. The alignment of increased rainfall with regions of significant negative vorticity advection reinforces the understanding that convergent motions are integral to developing and sustaining rain within monsoon depressions. This congruence between the observed rainfall and vorticity advection patterns substantiates the physical mechanisms driving monsoon depression dynamics.

Vertically Integrated Moisture Convergence (VIMC) is a crucial parameter that quantifies the net horizontal influx of moisture into a column of air, with positive values indicating convergence and potential for precipitation. The values of VIMC are high in the southwest sector for both monsoon depressions and monsoon lows, which implies a strong

convergence of moist air in this sector (Fig. 8b and e). This convergence is a process for the formation and intensification of rainfall, as it leads to the lifting and cooling of moist air, thereby promoting cloud formation and precipitation. In the case of monsoon depressions, the magnitude of VIMC is significantly high, which aligns with the rainfall patterns of both systems. The peak values of VIMC comparable well with the regions of highest rainfall observed in rainfall composites (Fig. 4a), confirming that areas with greater moisture convergence are indeed the same areas experiencing enhanced precipitation. This relationship between VIMC and rainfall patterns validates the observed precipitation composites and highlights the importance of moisture convergence in developing monsoon depressions. The central and southwestern sectors show a strong positive correlation, supporting the idea that regions with substantial moisture convergence are essential for the creation of precipitation (Fig. 9b and e). The high correlation between VIMC and precipitation highlights the significance of moisture accessibility in the deepening of monsoon depressions and lows, especially in these regions where moist air is abundantly supplied and lifted. The quantitative assessment of VIMC thus provides vital insights into the moisture dynamics that drive monsoon rainfall, underpinning the complex interaction between atmospheric moisture content and monsoon depression-induced precipitation.

Figure 8c and f shows the convergence of air and wind vectors around monsoon depression and monsoon lows centers throughout the study period. The noticeable

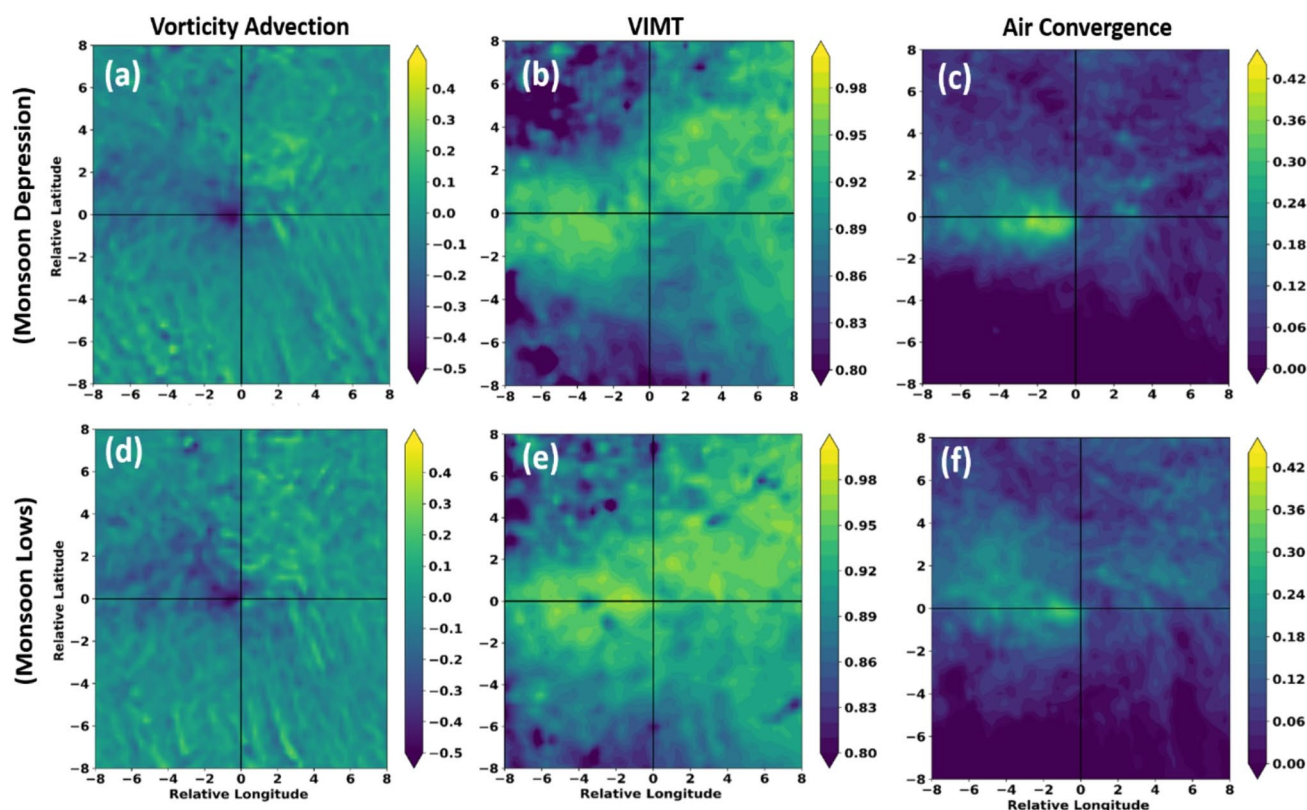


Fig. 9 Vortex-centered correlation maps of atmospheric variables correlated with rainfall (a) vorticity advection at 850 hPa, (b) vertically integrated moisture convergence from surface to 300 hPa, and (c) air

convergence associated with monsoon depressions at 850 hPa. (d-f) Correspond to the same as (a-c) but for monsoon lows

convergence, particularly in the central and southwestern sectors, aligns with regions of increased rainfall intensity in monsoon depressions (Fig. 4a). This indicates significant air convergence, essential for moisture elevation and subsequent cloud formation that results in precipitation during monsoon depressions. In the case of monsoon lows, the magnitude of convergence and wind speed is low (Fig. 8f). The circulation pattern is depicted by superimposed wind vectors, which validate convergent winds toward the depression's and lows' core, the location of the most significant rainfall. In addition to verified rainfall distribution, the wind flow and convergence patterns work together to provide moisture convergence in a dynamic context. The southwestern sector, while displaying a lesser degree of correlation than the central region in both monsoon depressions and lows, still indicates the relevance of convergent winds in precipitating processes (Fig. 9c and f). Understanding these correlations is essential for improving predictive models of monsoon rainfall, which are vital for effective water resource management and agricultural planning in regions affected by these systems. This analysis is pivotal for verifying the extent to which vorticity advection, VIMC, and air convergence can predict or explain the variability in rainfall during these monsoon systems.

4 Conclusion

The present study offers a detailed analysis of the spatial and temporal rainfall patterns associated with MLPs (i.e. monsoon depressions and monsoon lows) over the Indian subcontinent from 1979 to 2019 through ERA5 reanalysis data. Our findings not only reinforce but also expand upon the understanding presented in prior research by identifying specific patterns and trends in monsoon precipitation. Consistent with earlier studies (Vishnu et al. 2023), we observed a pronounced intensification of rainfall in southwestern quadrants. However this study brings new insights into the variability and increasing trend of rainfall, trends of MLPs, trends of mean positions of these systems, seasonal trends of rainfall, impact of ENSO on MLPs, and correlation of dynamic parameters with rainfall of these systems. This study also highlights all these shifts for monsoon lows, often less studied.

The initial examination of rainfall composites revealed a marked variability in precipitation distribution, with a significant increase in the southwestern quadrant, suggesting a favored sector for monsoon depression-induced rainfall. These findings are consistent across different phases of the monsoon season, with the most intense precipitation

occurring in June, diminishing by September, reflecting the lifecycle of the monsoon. The trend analysis suggests a clear difference between the rainfall trends of monsoon depression and lows. In both systems, the prominent quadrant for the increasing trend differs, a topic of further studies.

The study also looks into the climatic influences of ENSO events, where distinct contrasts were observed in rainfall patterns during the La Niña and El Niño years. La Niña conditions were associated with higher precipitation intensities within the MLPs' centers, while El Niño years showed relatively more muted rainfall patterns. This demonstrates the intricate link between oceanic phenomena and regional climatic variations. Further analysis of vorticity advection showed significant negative advection, particularly in the southwest sector, indicating strong convergence and validating the observed rainfall patterns. This physical mechanism was supported by the analysis of vertically integrated moisture convergence (VIMC), which presented a direct correlation with the enhanced precipitation in the same regions. This strengthened the idea that moisture convergence is a critical player in the development of rainfall linked to monsoon depressions.

This study provides a relationship between atmospheric dynamics and MLPs, highlighting their deep link. This interaction has significant implications, especially for local agriculture and hydrology. The study advances our knowledge of the climatic variables that influence the distribution and amount of rainfall associated with the monsoon. It also provides insightful information for strategic planning and predictive modeling in water resource management. The results also emphasize how important it is to continue observing and analyzing monsoon trends. This monitoring becomes even more critical in the face of potential climatic shifts. By doing so, we can better prepare for and mitigate the impacts of these crucial weather systems, especially in the densely populated regions of the Indian subcontinent.

Acknowledgements The authors are thankful to Dr. M. Mohapatra, Director General, India Meteorological Department, New Delhi for providing all facility and support to carry out this research work. We also acknowledge the ECMWF for providing reanalysis data used in this study.

Funding This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Declarations

Conflict of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

References

- Ajayamohan RS, Merryfield WJ, Kharin VV (2010) Increasing trend of synoptic activity and its relationship with extreme rain events over central India. *J Clim* 23:1004–1013. <https://doi.org/10.1175/2009JCLI2918.1>
- BHALME HN (1972) Trends and quasi-biennial Oscillation in the series of cyclonic disturbances over the Indian region. *MAUSAM* 23:355–358. <https://doi.org/10.54302/mausam.v23i3.5288>
- Bhatla R, Varma P, Verma S, Ghosh S (2020) El nino/la Nina impact on crop production over different agro-climatic zones of Indo-Gangetic plain of India. *Theor Appl Climatol* 142:151–163. <https://doi.org/10.1007/S00704-020-03284-3>
- Boos WR, Kuang Z (2010) Dominant control of the South Asian monsoon by orographic insulation versus plateau heating. *Nature* 463:218–222. <https://doi.org/10.1038/NATURE08707>
- Chakraborty A, Nanjundiah RS, Srinivasan J (2002) Role of Asian and African orography in Indian summer monsoon. *Geophys Res Lett* 29. <https://doi.org/10.1029/2002GL015522>
- Chakraborty A, Nanjundiah RS, Srinivasan J (2006) Theoretical aspects of the onset of Indian summer monsoon from perturbed orography simulations in a GCM. *Ann Geophys* 24:2075–2089. <https://doi.org/10.5194/angeo-24-2075-2006>
- Dash SK, Kumar JR, Shekhar MS (2004) On the decreasing frequency of monsoon depressions over the Indian region. *Curr Sci* 86(10):1404–1411
- Ding Y, Sikka DR (2006) Synoptic systems and weather. *Asian Monsoon* 131–201. https://doi.org/10.1007/3-540-37722-0_4
- Goswami BN (1987) A mechanism for the west-north-west movement of monsoon depressions. *Nature* 326:376–378. <https://doi.org/10.1038/326376a0>
- Hahn DG, Manabe S (1975) The role of mountains in the south Asian monsoon circulation. *J Atmos Sci* 32(8):1515–1541
- Hersbach H, Bell B, Berrisford P et al (2020) The ERA5 global reanalysis. *Q J R Meteorol Soc* 146:1999–2049. <https://doi.org/10.1002/qj.3803>
- Hunt KMR, Fletcher JK (2019) The relationship between Indian monsoon rainfall and low-pressure systems. *Clim Dyn* 53:1859–1871. <https://doi.org/10.1007/s00382-019-04744-x>
- Hurley JV, Boos WR (2015) A global climatology of monsoon low-pressure systems. *Q J R Meteorol Soc* 141:1049–1064. <https://doi.org/10.1002/qj.2447>
- Kalnay E, Collins W, Deaven D et al (1996) The NCEP NCAR 40-year reanalysis project. 1996.pdf. *Bull Am Meteorol Soc* 77:437–472
- Keshavamurty RN, Satyan V, Goswami BN (1978) Indian summer monsoon cyclogenesis and its variability [10]. *Nature* 274:576–578
- KESHAVAMURTY RN (1972) Certain aspects of monsoon depressions as revealed problems of by satellite pictures. *MAUSAM* 23:161–164. <https://doi.org/10.54302/mausam.v23i2.5271>
- Lal M, Nozawa T, Emori S, Harasawa H, Takahashi K, Kimoto M, Abe-Ouchi A, Nakajima T, Takemura TU, Numaguti A (2001) Future climate change: implications for Indian summer monsoon and its variability. *Curr Sci* 81(9):1196–1207
- Lohan N, Routray A, Kumar R, Mahala BK, Kumar S (2025) Validation and diagnostic study of cloudburst events over the Himalayan region using IMDAA and ERA5. *Geomatics, Natural Hazards and Risk* 16(1):2446589
- Martin JE (2014) Quasi-geostrophic diagnosis of the influence of vorticity advection on the development of upper level jet-front systems. *Q J R Meteorol Soc* 140:2658–2671. <https://doi.org/10.1002/qj.2333>
- Mohapatra M, Mohanty UC (2009) Excess and deficient summer monsoon rainfall over Orissa in relation to low pressure systems. *Mausam* 60:25–38. <https://doi.org/10.54302/mausam.v60i1.959>

- Mohapatra M, Srivastava AK, Balachandran S et al (2017) Inter-annual variation and trends in tropical cyclones and monsoon depressions over the North Indian Ocean. In: Rajeevan MN, Nayak S (eds) *Observed climate variability and change over the Indian region*. Springer, Singapore, pp 89–106. https://doi.org/10.1007/978-981-10-2531-0_6
- Mondal A, Kundu S, Mukhopadhyay A (2012) Rainfall trend analysis by Mann-Kendall test: A case study of north-eastern part of cuttack district, orissa, international journal of geology. *Earth Environ Sci* 1:70–78
- MOOLEY D, SHUKLA J (1989) Main features of the westward-moving low pressure systems which form over the Indian region during the summer monsoon season and their relation to the monsoon rainfall. *MAUSAM* 40:35–50. <https://doi.org/10.54302/mausam.v40i2.2041>
- Nie J, Boos WR, Kuang Z (2010) Observational evaluation of a convective quasi-equilibrium view of monsoons. *J Clim* 23:4416–4428. <https://doi.org/10.1175/2010JCLI3505.1>
- Nikumbh AC, Chakraborty A, Bhat GS, Frierson DMW (2020) Large-Scale extreme Rainfall-Producing synoptic systems of the Indian summer monsoon. *Geophys Res Lett* 47. <https://doi.org/10.1029/2020GL088403>
- Parthasarathy B, Munot AA, Kothawale DR (1994) All-India monthly and seasonal rainfall series: 1871–1993. *Theor Appl Climatol* 49:217–224. <https://doi.org/10.1007/BF00867461>
- Pattanaik DR (2007) Analysis of rainfall over different homogeneous regions of India in relation to variability in westward movement frequency of monsoon depressions. In: *Natural Hazards*. pp 635–646
- Pattanaik DR, Rajeevan M (2010) Variability of extreme rainfall events over India during Southwest monsoon season. *Meteorol Appl* 17:88–104. <https://doi.org/10.1002/met.164>
- RAO YP (1961) Some characteristics of the Southwest monsoon circulation. *MAUSAM* 12:413–418. <https://doi.org/10.54302/MAUSAM.V12I3.4213>
- Sawyer JS, Matthewman AG (1951) On the evaluation of terms of a type arising in Sutcliffe's treatment of cyclonic development. *Q J R Meteorol Soc* 77:667–671. <https://doi.org/10.1002/qj.49707733411>
- Sikka DR (1977) Some aspects of the life history, structure and movement of monsoon depressions. *Pure Appl Geophys PAGEOPH* 115:1501–1529. <https://doi.org/10.1007/BF00874421>
- Thomas TM, Bala G, Srinivas VV (2021) Characteristics of the monsoon low pressure systems in the Indian Subcontinent and the associated extreme precipitation events. *Clim Dyn* 56:1859–1878. <https://doi.org/10.1007/s00382-020-05562-2>
- Thomas TM, Bala G, Vemavarapu SV, Boos WR (2023) Opposite changes in monsoon precipitation and low pressure system frequency in response to orographic forcing. *J Clim* 36:6309–6329. <https://doi.org/10.1175/JCLI-D-22-0476.1>
- Ullah K, Gao S (2012) Moisture transport over the Arabian sea associated with summer rainfall over Pakistan in 1994 and 2002. *Adv Atmos Sci* 29:501–508. <https://doi.org/10.1007/S00376-011-0200-Y/METRICS>
- Vishnu S, Boos WR, Ullrich PA, O'Brien TA (2020) Assessing historical variability of South Asian monsoon lows and depressions with an optimized tracking algorithm. *J Geophys Res: Atmos* 125(15):e2020JD032977. <https://doi.org/10.1029/2020JD032977>
- Vishnu S, Risser MD, O'Brien TA et al (2023) Observed increase in the peak rain rates of monsoon depressions. *Npj Clim Atmos Sci* 6. <https://doi.org/10.1038/s41612-023-00436-w>
- Zhang R, Jiang D, Zhang Z, Yu E (2015) The impact of regional uplift of the Tibetan plateau on the Asian monsoon climate. *Palaeogeogr Palaeoclimatol Palaeoecol* 417:137–150. <https://doi.org/10.1016/J.PALAEO.2014.10.030>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.